

Curriculum Vitae – Ambra Ferrari, PhD

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Professional Experience

01/05/2026 – present	Assistant Professor (Tenure-Track Researcher)
Affiliation	SISSA – Scuola Internazionale Superiore di Studi Avanzati, Trieste, Italy
01/05/2024 – 30/04/2026	Marie Skłodowska-Curie Postdoctoral Fellow
Affiliation	CIMeC – Center for Mind/Brain Sciences, University of Trento, Rovereto, Italy
Project	Multimodal language in preverbal infancy (ERP and multivariate decoding of infant and adult EEG)
Supervisor	Prof. Eugenio Parise
01/08/2022 – 30/04/2024	Postdoctoral Researcher
Affiliation	Max Plank Institute for Psycholinguistics & Donders Institute for Brain, Cognition and Behaviour, Radboud University, Nijmegen, The Netherlands
Project	Multimodal integration and prediction in face-to-face communication (online audiovisual testing, psycholinguistics, Bayesian linear mixed modelling)
Supervisor	Prof. Peter Hagoort
01/01/2020 – 31/07/2022	Postdoctoral Researcher
Affiliation	Donders Institute for Brain, Cognition and Behaviour, Radboud University, Nijmegen, The Netherlands
Project	Statistical learning and predictive processing in the human brain (online testing, signal detection theory, univariate and multivariate fMRI)
Supervisor	Prof. Floris P. de Lange

Education

23/06/2021	Professional Psychology Qualification (Abilitazione Professionale)
	National examination to access the Italian register of psychologists (Albo) Passing Grade: 50/50
07/09/2015 – 31/12/2019	PhD in Computational Cognitive Neuroscience
Affiliation	Computational Cognitive Neuroimaging Group, School of Psychology, University of Birmingham, Birmingham, United Kingdom
Thesis title	The interplay between attention and multisensory integration (psychophysics, Bayesian computational modelling, univariate and multivariate fMRI and EEG)
Supervisor	Prof. Uta Noppeney
15/09/2014 – 01/09/2015	Apprenticeship for National Psychology Certification
Affiliation	G.B. Rossi Hospital, Verona; Sant'Agostino Estense Hospital, Modena, Italy
Activity	Clinical neuropsychology assessment and rehabilitation of cognitive disorders following neural damage; cognitive stimulation and monitoring in Alzheimer's Unit; fMRI mapping of linguistic and motor brain areas prior to neurosurgery
Supervisors	Dr. Valentina Varalta; Dr. Francesca Benuzzi
09/09/2014	Master's Degree in Cognitive Neuropsychology and Neuroscience
	Passing Grade: 110/110 (cum laude)
Affiliation	School of Cognitive Science, University of Trento, Rovereto, Italy
Thesis title	Investigating cross-modal plasticity in deafness: an fMRI pilot study on global and biological motion perception
Supervisor	Prof. Olivier Collignon
11/09/2012	Bachelor's Degree in Cognitive Psychology
	Passing Grade: 110/110 (cum laude)
Affiliation	School of Cognitive Science, University of Trento, Rovereto, Italy

Thesis title Use of a light source for spatial reorientation by the species of fish *Danio Rerio*
 Supervisor Prof. Valeria Anna Sovrano

Teaching and Supervision

01/05/2026 – present	Supervisor of graduate students
Affiliation	SISSA – Scuola Internazionale Superiore di Studi Avanzati, Trieste, Italy
Activity	Weekly supervision of 2 PhD students for experimental design, data acquisition, data analysis and manuscript preparation (techniques: psychophysics; MRI; EEG)
01/05/2024 – 30/04/2026	Guest Lecturer
Affiliation	Department of Psychology and Cognitive Science, University of Trento
Activity	Cognitive Neuroscience (Prof. Francesco Pavani) Content: frontal teaching on the computational cognitive neuroscience of multisensory integration (computation, algorithm, implementation)
Affiliation	CIMeC – Center for Mind/Brain Sciences, University of Trento
Activity	Cognitive Neuroscience of Infant Development (Prof. Eugenio Parise) Content: interactive teaching (Journal Club format) on language development (semantics, pragmatics) in preverbal infancy
01/05/2024 – present	Co-supervisor of undergraduate and graduate students
Affiliation	CIMeC – Center for Mind/Brain Sciences, University of Trento, Rovereto, Italy
Activity	Weekly co-supervision of 1 PhD student and 2 MSc students for experimental design, data acquisition, data analysis and manuscript preparation (techniques: psychophysics; MRI; infant and adult EEG)
01/09/2020 – 30/04/2024	Co-supervisor of undergraduate and graduate students
Affiliation	Max Plank Institute for Psycholinguistics & Donders Institute for Brain, Cognition and Behaviour, Radboud University, Nijmegen, The Netherlands
Activity	Weekly co-supervision of 2 PhD students for experimental design, data acquisition, data analysis and manuscript preparation (techniques: online testing; psychophysics; MRI); organization and supervision of training activities for 3 lab rotation MSc students
01/10/2017 – 31/07/2018	Co-supervisor of undergraduate students
Affiliation	School of Psychology, University of Birmingham, United Kingdom
Activity	Co-supervision of 1 BSc student for research internship and dissertation (techniques: psychophysics; electrodermal activity and pupillometry recordings)
01/10/2016 – 31/07/2017	Teaching assistant
Affiliation	School of Psychology, University of Birmingham, United Kingdom
Activity	Delivery of practical workshops on statistics for BSc students
01/10/2013 – 31/07/2014	Teaching assistant
Affiliation	School of Cognitive Science, University of Trento, Rovereto, Italy
Activity	Delivery of practical workshops on statistics for BSc students

Academic Service

08/2023 – present	Review Editor
	Frontiers in Psychiatry – Section Social Neuroscience
12/2021 – present	Ad-hoc reviewer
Activity	Peer-review in Q1 journals (eLife; Cereb Cortex; Imaging Neurosci; J Cog Neurosci; Communications Biology; Developmental Cog Neurosci; Psych Science; Communications Psychology; Cognition; Neurosci Biobehav Rev; Ann NY Acad Sci; Eur J Neurosci; Psychon Bull Rev; Neuropsychologia; Plos

One; Front Neurosci; Front Hum Neurosci; J Exp Psy: HPP; Q J Exp Psy; Mult Res; Lang Cogn Neurosci; Brain & Language)

2025 – present **Member of PhD admission committee**
SISSA Cognitive Neuroscience PhD program (Italy)

2022 – present **Examiner of PhD evaluation committee**
2025: SISSA (Italy); 2024: Sapienza University (Italy); 2022: CIMEC, University of Trento (Italy)

01/10/2021 – 30/09/2022 **Postdoctoral representative**
Affiliation Donders Centre for Cognitive Neuroimaging, Donders Institute for Brain, Cognition and Behaviour, Radboud University, Nijmegen, The Netherlands
Activity Onboarding of new postdocs; organization of Brown Bag Meetings; co-organization of Donders Career Day 2021; organization of postdocs social activities; organization of postdoc information centre on the institute intranet

01/09/2021 – 31/07/2022 **Journal Club organizer and chair**
Affiliation Predictive Brain Lab, Donders Institute for Brain, Cognition and Behaviour, Radboud University, Nijmegen, The Netherlands
Activity Organization of annual calendar and chairing of biweekly meetings

Publications and Manuscripts

Peer-reviewed articles and preprints

Ferrari, A., de Lange, F. P., Akrami, A. (2026). Where learning paths meet: Convergence and divergence of statistical and reinforcement learning. *Current Opinion in Neurobiology*, 98, 103181.

Degano, G.*, **Ferrari, A.***, Noppeney, U. (2026). Feeling the music: Audiotactile encoding of temporal structure in the human brain. *bioRxiv*. ***shared first authorship**

Mazzi, G.*, **Ferrari, A.***, Mencaroni, M. L., Valzolgher, C., Tommasini, M., Pavani, F., Benetti, S. (2025). Prior expectations guide multisensory integration during face-to-face communication. *PLoS Computational Biology*, 21(9), e1013468. ***shared first authorship**

Ferrari, A., Filippin, L., Buiatti, M., Parise, E. (2025). WTools: a MATLAB-based toolbox for time-frequency analysis of infant data. *PLoS One*, 20(5), e0323179.

Momentè, F., Suglia, A., Giulianelli, M., **Ferrari, A.**, Koller, A., Lemon, O., Schlangen, D., Fernández, R., Bernardi, R. (2025). Triangulating LLM progress through benchmarks, games, and cognitive tests. *arXiv*.

Ferrari, A., Hagoort, P. (2025). Beat gestures and prosodic prominence interactively influence language comprehension. *Cognition*, 256, 106049.

Nazli, I., **Ferrari, A.**, Huber-Huber, C., de Lange, F. P. (2024). Forward and backward blocking in statistical learning. *PLoS One*, 19(8), e0306797.

Benetti, S.*, **Ferrari, A.***, Pavani, F. (2023). Multimodal processing in face-to-face interactions: a bridging link between psycholinguistics and sensory neuroscience. *Frontiers in Human Neuroscience*. 17:1108354. ***shared first authorship**

Ferrari, A., Richter, D., de Lange, F. (2022). Updating contextual sensory expectations for adaptive behaviour. *The Journal of Neuroscience*, 42(47), 8855-8869.

Ferrari, A., Noppeney, U. (2021). Attention controls multisensory perception via two distinct mechanisms at different levels of the cortical hierarchy. *PLoS biology*, 19(11), e3001465.

Benetti, S., Zonca, J., **Ferrari, A.**, Rezk, M., Rabini, G., Collignon, O. (2021). Visual motion processing recruits regions selective for auditory motion in early deaf individuals. *NeuroImage*, 230, 117816.

Hickman, L. J., Keating, C. T., **Ferrari, A.**, Cook, J. L. (2021). Skin conductance as an index of alexithymic traits in the general population. *Psychological Reports*, 00332941211005118.

Beierholm, U., Rohe, T., **Ferrari, A.**, Stegle, O., Noppeney, U. (2020). Using the past to estimate sensory uncertainty. *eLife*, 9, e54172.

Noppeney, U., Jones, S. A., Rohe, T., **Ferrari, A.** (2018). See what you hear - How the human brain forms representations across the senses. *Neuroforum*, A169-A181.

Lee, S. A., **Ferrari, A.**, Vallortigara, G., Sovrano, V. A. (2015). Boundary primacy in spatial mapping: Evidence from zebrafish (*Danio rerio*). *Behavioural processes*, 119, 116-122.

Manuscripts in preparation

Ferrari, A.*, Mazzi, G.*, Rabini, G., Mencaroni, M. L., Pavani, F., Benetti, S. Binding voices and faces: How the brain controls multisensory integration during face-to-face communication.

Ferrari, A., Meijer, D., Noppeney, U. Attention selection and switching in multisensory causal inference.

Invited talks

Ferrari, A. (2026). How the brain controls multisensory perception. Invited by AWARENET Team, Padova Neuroscience Center, Italy

Ferrari, A. (2025). How the brain merges information across the senses. Invited Talk at the Italian Network for Computational and Theoretical Neuroscience, Padova, Italy

Ferrari, A. (2025). The Bayesian mind: perceiving the world is an inference problem. Invited by Prof. Marco Formentin, Padova Neuroscience Center, Italy

Ferrari, A. (2024). Multimodality as a design feature of human cognition. Invited by Cognitive Neuroscience Department, SISSA, Italy

Ferrari, A. (2024). Bayesian Causal Inference in multimodal communication. Invited by Prof. Gabriella Vigliocco, UCL, UK

Ferrari, A. (2022). Bayesian Causal Inference in the mind and brain. Invited by Prof. Francesco Pavani, CIMEC, University of Trento, Italy

Conference Symposium Organization

Chairing of Nanosymposium "Cross-modal processing in humans" at SfN Neuroscience 2024, Chicago, IL, USA. 05/10/2024 – 09/10/2024.

Co-organization and chairing of the symposium "Multisensory processing in face-to-face communication" at the 21st International Multisensory Research Forum, Brussels, Belgium. 27/06/2023 – 30/06/2023

Organization and chairing of the symposium "How do we learn and predict structure in the world?" at the 18th NVP Winter Conference on Brain and Cognition, Egmond aan Zee, The Netherlands. 28/04/2022 – 30/04/2022

Conference Oral Presentations

Ferrari, A. (2025). Binding voices and faces: How the brain uses prior expectations to integrate multisensory signals in communication. Talk presented at the 23rd International Multisensory Research Forum, Durham, UK. 15/07/2025 – 18/07/2025

Ferrari, A. (2024). Audiotactile encoding of music in the human brain. Nanosymposium talk at SfN Neuroscience 2024, Chicago, IL (USA). 05/10/2024 – 09/10/2024

Ferrari A. et al. (2024). Statistical learning of a hierarchical centre-embedded structure: influence of Zipf distribution and semantic biases. Talk presented at the Interdisciplinary Advances in Statistical Learning Conference, San Sebastián, Basque Country, Spain. 05/06/2024 – 07/06/2024

Ferrari, A. (2023). Interactive effects of prosody and beat gestures on spoken language comprehension. Talk presented at the 19th NVP Winter Conference on Brain and Cognition, Egmond aan Zee, The Netherlands. 14/12/2023 – 16/12/2023

Ferrari, A. (2023). Towards a neurocognitive model of multisensory processing in face-to-face communication. Talk presented at the 21st International Multisensory Research Forum, Brussels, Belgium. 27/06/2023 – 30/06/2023

Ferrari, A. (2022). Updating contextual visual expectations. Talk presented at the European Conference on Visual Perception (ECVP 2022), Nijmegen, The Netherlands. 28/08/2022 – 01/09/2022

Ferrari, A. et al. (2022). The neural signatures of incidental and intentional statistical learning. Talk presented at the 18th NVP Winter Conference on Brain and Cognition, Egmond aan Zee, The Netherlands. 28/04/2022 – 30/04/2022

Ferrari, A. (2019). Neural mechanisms of attention in multisensory decision-making. Talk presented at the Oscillations Workshop, Centre for Human Brain Health, Birmingham, UK. 10/05/2019

Ferrari, A. (2018). The interplay between attention and multisensory integration. Talk presented at the Attention & Oscillations Workshop, Centre for Human Brain Health, Birmingham, UK. 16/11/2018

Conference Poster Presentations

Ondei, D., Rabini, G., **Ferrari, A.**, Trujillo, J., Holler, J., Pavani, F., Benetti, S. (2026). From face and voice to intent: behavioral signatures of audiovisual integration in face-to-face communication. Poster presented at the Concepts, Actions and Objects (CAOs) Workshop, Rovereto, Italy. 06/05/2025 – 08/05/2025

Ferrari, A., Rabini, G., Trujillo, J., Mencaroni, M. L., Holler, J., Pavani, F., Benetti, S. (2026). From signals to intentions: neural mechanisms of audiovisual integration in face-to-face communication. Poster presented at the Concepts, Actions and Objects (CAOs) Workshop, Rovereto, Italy. 06/05/2025 – 08/05/2025

Ferrari, A., Mazzi, G., Rabini, G., Mencaroni, M. L., Tommasini, M., Valzolgher, C., Pavani, F., Benetti, S. (2025). How the brain controls multisensory integration in social action processing. Poster presented at the Concepts, Actions and Objects (CAOs) Workshop, Rovereto, Italy. 07/05/2025 – 09/05/2025

Ferrari, A., Hagoort, P. (2024). Beat gestures and prosodic stress interactively influence language comprehension. Poster presented at the 28th Workshop on the Semantics and Pragmatics of Dialogue, Rovereto, Italy. 11/09/2024 – 12/09/2024

Ferrari, A., Mazzi, G., Mencaroni, M. L., Valzolgher, C., Pavani, F., Benetti, S. (2024). Bayesian Causal Inference governs multisensory integration during face-to-face communication. Poster presented at the Highlights in the Language Sciences Conference, Nijmegen, The Netherlands. 08/07/2024 – 11/07/2024

Mazzi, G., Mencaroni, M. L., Rabini, G., Tommasini, M., Valzolgher, C., Pavani, F., **Ferrari, A.**, Benetti, S. (2024). Face-to-face social interaction in the human brain: assessing the role of a lateral processing pathway. Poster presented at the Concepts, Actions and Objects (CAOs) Workshop, Rovereto, Italy. 09/05/2024 – 11/05/2024

Ferrari, A., Mazzi, G., Mencaroni, M. L., Valzolgher, C., Pavani, F., Benetti, S. (2024). Prior expectations guide multisensory integration during face-to-face communication. Poster presented at the Concepts, Actions and Objects (CAOs) Workshop, Rovereto, Italy. 09/05/2024 – 11/05/2024

Chen Y., **Ferrari A.**, Hagoort, P., Bocanegra, B., Poletiek, F. (2023). Learning hierarchical centre-embedded structures: influence of distributional properties of the input. Poster presented at the 19th NVP Winter Conference on Brain and Cognition, Egmond aan Zee, The Netherlands. 14/12/2023 – 16/12/2023

Ferrari, A. (2023). How does audiovisual prosody influence spoken language comprehension? Poster presented at the 15th Annual Meeting of the Society for the Neurobiology of Language, Marseille, France. 24/10/2023 – 26/10/2023

Mazzi, G., **Ferrari, A.**, Valzolgher, C., Tommasini, M., Pavani, F., Benetti, S. (2023). Domain-general Bayesian causal inference in multisensory processing of face-to-face interactions. Poster presented at the Concepts, Actions and Objects (CAOs) Workshop, Rovereto, Italy. 04/05/2023 – 06/05/2023

Nazli, I., **Ferrari, A.**, Huber-Huber, C., de Lange, F. P. (2022). Is statistical learning error-driven? Poster presented at the 4th International Conference on Interdisciplinary Advances in Statistical Learning, San Sebastian, Spain. 01/06/2022 – 03/09/2022

Nazli, I., **Ferrari, A.**, Huber-Huber, C., de Lange, F. P. (2022). Is statistical learning error-driven?. Poster presented at the 18th NVP Winter Conference on Brain and Cognition, Egmond aan Zee, The Netherlands. 28/04/2022 – 30/04/2022

Ferrari, A., Noppeney, U. (2019). Attention influences how the brain integrates audiovisual signals into spatial representations. Poster presented at the 25th Annual Meeting of the Organization for Human Brain Mapping (OHBM 2019), Rome, Italy. 09/06/2019 – 13/06/2019

Benetti, S., Zonca, J., Rabini, G., **Ferrari, A.**, Foa, V., Rezk, M., Collignon, O. (2017). Functional selectivity for visual radial motion processing in the temporal auditory cortex of early deaf individuals. Poster presented at the Ten Years of Mind/Brain Sciences at the University of Trento, Rovereto, Italy. 19/10/2017 – 21/10/2017

Ferrari, A., Noppeney, U. (2017). Endogenous modality-specific attention influences sensory reliability during multisensory integration. Poster presented at the Ten Years of Mind/Brain Sciences at the University of Trento, Rovereto, Italy. 19/10/2017 – 21/10/2017

Ferrari, A., Noppeney, U. (2017). The role of endogenous modality-specific attention in multisensory integration. Poster presented at the International Multisensory Research Forum (IMRF 2017), Nashville, TN, USA. 19/05/2017 – 22/05/2017

Ferrari, A., Noppeney, U. (2017). The role of endogenous modality-specific attention in multisensory integration. Poster presented at the Festival of Neuroscience, British Neuroscience Association (BNA), Birmingham, UK. 10/04/2017 – 13/04/2017

Major Collaborations

Uta Noppeney, Donders Institute for Brain, Cognition and Behaviour, Radboud University, The Netherlands. Topic: Multisensory integration

Francesco Pavani, **Stefania Benetti**, **Eugenio Parise**, CIMEC, University of Trento, Italy; **Judith Holler**, **Linda Drijvers**, Max Plank Institute for Psycholinguistics, The Netherlands; **James P. Trujillo**, University of Amsterdam, The Netherlands. Topic: Language

Floris P. de Lange, Donders Institute for Brain, Cognition and Behaviour, Radboud University, The Netherlands; **Athena Akrami**, University College London, UK. Topic: Statistical learning and predictive processing

Grants & Awards

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| 2024 | SfN Training and Professional Development Award (\$2500)
Scholarship awarded based on outstanding academic profile to attend SfN Neuroscience conference in Chicago, USA |
| 2023 | Marie Skłodowska-Curie European Postdoctoral Fellowship (€189.000)
Multimodal meanings for conceptual communication before and beyond spoken language (MultiMeans) |
| 2019 | Guarantors of Brain Travel Grant (£500)
Financial support to attend OHBM conference in Rome, Italy |
| 2016 | University of Trento Merit Prize (€2,700)
Scholarship awarded for outstanding achievement during MSc course |
| 2014 | University of Trento Merit Prize (€2,700)
Scholarship awarded for outstanding achievement during BSc course |

Communication and Outreach

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| 26/11/2025 | Guest speaker at “Chiacchiere in mescita” podcast
Invited interview on the cognitive neuroscience and psycholinguistics of non-verbal communication in preverbal infants |
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11/09/2024 Activity	Article in UniTrento Magazine Outreach article titled “Prima delle parole” to present and promote my research activities to the UniTrento community and beyond
29/11/2023	Blog post in Het Talige Brein Outreach article titled “How can we communicate with babies?” to share my passion for the study of multimodal communication from the first years of life
18-19/03/2023 Place Activity	Volunteer at InScience Film Festival Nijmegen, The Netherlands Accreditations and info desk management during annual science film festival
28/11/2022 Activity	Guest speaker at Scientificast podcast Invited interview on the cognitive neuroscience of multimodal integration and learning by Italian top-chart science podcast
23/09/2021 Activity	Guest speaker at Youtube channel for science outreach Invited live talk on the cognitive neuroscience of learning by Italian Youtube channel dedicated to science dissemination and outreach (IlariaFMATH)
2017 – 2019 Place Activity	Pint of Science event manager Birmingham, United Kingdom Organization of 3 editions of an annual science festival (create and manage team of 5 volunteers; book venue and 6 speakers; organize and deliver science demos (e.g. perceptual illusions); host event
2016 – 2019 Place Activity	Open days volunteer School of Psychology, University of Birmingham, United Kingdom Laboratory demonstrations (piezo-tactile stimulation; eye tracking) during university open days for high school students and their families

Skills

Trainings	Donders Institute Advanced M/EEG Toolkit 2025: 5-day intensive course on advanced M/EEG data analysis (pre-processing, ERP, frequency analysis, source reconstruction, TRFs, statistical methods) Donders Centre for Cognitive Neuroimaging (Nijmegen, NL): 3T Siemens MRI scanner operator qualification (2021: ~80h scanning) Centre for Human Brain Health (Birmingham, UK): 3T Philips MRI scanner operator qualification (2017-2019: ~150h scanning)
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Technical Knowledge	Program	Area	Knowledge
	MATLAB	Programming language	Advanced
	Python	Programming language	Advanced
	Jupyter Notebook	Python coding Interface	Advanced
	SPM	MATLAB MRI data analysis	Advanced
	FSL	Python MRI data analysis	Good
	The Decoding Toolbox	MATLAB MRI data analysis	Advanced
	FreeSurfer	Python MRI data analysis	Good
	WTools	MATLAB EEG data analysis	Advanced
	SPSS	Data analysis	Good
	Bambi	Python Bayesian modelling	Advanced
	Eyelink	Eye-tracking data acquisition	Advanced
	BIOPAC Student Lab	Physiological data acquisition	Basic
	PsPM	Psycho-physiological data analysis	Basic
	Psychtoolbox	MATLAB experiment builder	Advanced
	NBS Presentation	Experiment builder	Advanced
	Gorilla	Online experiment builder	Advanced
	Academic Prolific	Online experiment recruitment	Advanced

PsychoPy
Inkscape

Python experiment builder
Vector graphics editor

Basic
Good

**Creative and networking
skills**

04/2021 – 06/2022: Creator and host of joint group meetings for the Predictive Brain lab (PI: Floris P. de Lange). Tasks: design structure of event; establish connection with other labs; organize panel of speakers across labs; host meetings. Established network of PIs: Lucia Melloni, Lars Muckli, Jan Theeuwes, Athena Akrami, Stefan van der Stigchel.

07/10/2019: Creator and host of academic skills workshop “What Next? Career tips for PhD students” at the School of Psychology, University of Birmingham, UK. Tasks: design structure of event; organize panel of speakers; organize refreshments through university funds; host meetings.

Languages

Italian (native speaker); **English** (advanced user, IELTS C1 level)

Last modified: 18.05.2026